

☒ Easy (10 Questions)

1.

What is the main goal of phishing attacks?

- A) Steal user credentials
- B) Improve network speed
- C) Encrypt data
- D) Update software

Answer: A

Explanation: Phishing attacks aim to trick users into revealing sensitive information such as usernames and passwords.

2.

Which Android component is commonly exploited in web-based attacks?

- A) WebView
- B) Camera
- C) GPS
- D) Bluetooth

Answer: A

Explanation: WebView renders web content inside Android apps, making it a common attack vector if improperly secured.

3.

What is a Man-in-the-Middle (MITM) attack?

- A) Intercepting communication between two parties
- B) Installing an app from the Play Store
- C) Updating system software
- D) Encrypting user data

Answer: A

Explanation: MITM attacks intercept or modify communication between two parties without their knowledge.

4.

What type of network does an attacker often exploit for MITM attacks?

- A) Public Wi-Fi
- B) Cellular network
- C) Bluetooth connection
- D) VPN connection

Answer: A

Explanation: Public Wi-Fi is often unsecured and easily intercepted by attackers for MITM.

5.

Which of these is a common social engineering attack?

- A) Phishing
- B) SQL Injection
- C) Buffer Overflow
- D) Malware encryption

Answer: A

Explanation: Phishing manipulates people into revealing confidential info, a typical social engineering tactic.

6.

What does ARP poisoning attack target?

- A) Network traffic
- B) Battery life
- C) Screen brightness
- D) Bluetooth settings

Answer: A

Explanation: ARP poisoning manipulates the Address Resolution Protocol to intercept network traffic.

7.

Which of these is an example of a web-based attack on Android?

- A) Cross-Site Scripting (XSS)

- B) Bluetooth spoofing
- C) GPS tracking
- D) App sandboxing

Answer: A

Explanation: XSS injects malicious scripts into web content, often exploited via WebViews.

8.

What does a rogue access point do?

- A) Pretends to be a legitimate Wi-Fi hotspot
- B) Boosts network speed
- C) Updates apps automatically
- D) Blocks malware

Answer: A

Explanation: Rogue APs trick users into connecting to fake hotspots to intercept data.

9.

Which Android feature can be exploited by malicious apps to perform Man-in-the-Browser attacks?

- A) Accessibility permissions
- B) Battery saver
- C) Camera flash
- D) Screen lock

Answer: A

Explanation: Accessibility permissions can allow apps to intercept or manipulate user input.

10.

What is the main purpose of input validation in apps?

- A) Prevent malicious data injection
- B) Increase battery usage
- C) Improve UI responsiveness
- D) Speed up downloads

Answer: A

Explanation: Input validation ensures only safe and correct data is processed, preventing attacks like XSS or injection.

☒ Medium (20 Questions)

11.

What type of attack uses fake Wi-Fi hotspots to intercept data?

- A) Evil Twin Attack
- B) Phishing
- C) SQL Injection
- D) Buffer Overflow

Answer: A

Explanation: Evil Twin attacks create rogue access points mimicking legitimate Wi-Fi to capture user traffic.

12.

Which HTTP header helps prevent Clickjacking attacks?

- A) X-Frame-Options
- B) Content-Type
- C) Cache-Control
- D) User-Agent

Answer: A

Explanation: X-Frame-Options restricts how browsers render content in frames, blocking clickjacking.

13.

What is DNS spoofing?

- A) Redirecting DNS requests to malicious IP addresses
- B) Encrypting DNS traffic
- C) Blocking DNS queries
- D) Updating DNS servers automatically

Answer: A

Explanation: DNS spoofing alters DNS responses to redirect users to attacker-controlled sites.

14.

Which Burp Suite tool automates attacking input fields?

- A) Intruder
- B) Repeater
- C) Spider
- D) Decoder

Answer: A

Explanation: Intruder performs automated attacks like brute force or fuzzing on inputs.

15.

How can apps defend against Man-in-the-Middle attacks?

- A) Use SSL/TLS with certificate pinning
- B) Disable Wi-Fi
- C) Turn off GPS
- D) Use VPN only on desktops

Answer: A

Explanation: Certificate pinning prevents attackers from intercepting TLS connections with fake certificates.

16.

What is a common sign of a Trojan infection on Android?

- A) Unexpected app behavior or permissions
- B) Faster charging
- C) Improved screen brightness

D) Better network speed

Answer: A

Explanation: Trojans often request unusual permissions or behave suspiciously.

17.

What is 'smishing'?

A) Phishing via SMS messages

B) Phishing via email

C) Network sniffing

D) Bluetooth hacking

Answer: A

Explanation: Smishing uses SMS to trick users into malicious actions.

18.

Which of these is NOT a type of port scan?

A) NULL scan

B) SYN scan

C) FIN scan

D) SSL scan

Answer: D

Explanation: SSL scan is not a port scan type; SYN, FIN, and NULL are common scan techniques.

19.

What does a 'drive-by download' mean?

A) Downloading malware automatically upon visiting a website

B) Downloading apps manually

C) Uploading files

D) Backing up data

Answer: A

Explanation: Drive-by downloads install malware without user consent when visiting malicious sites.

20.

Which social engineering technique involves impersonating a trusted person over phone?

- A) Vishing
- B) Smishing
- C) Phishing
- D) Tailgating

Answer: A

Explanation: Vishing is voice phishing over calls.

21.

Which network attack involves poisoning the ARP cache?

- A) ARP Spoofing
- B) DNS Spoofing
- C) SYN Flood
- D) Phishing

Answer: A

Explanation: ARP spoofing tricks devices into sending data to attacker-controlled MAC addresses.

22.

What Android permission is critical to monitor for preventing malicious overlay attacks?

- A) SYSTEM_ALERT_WINDOW
- B) INTERNET
- C) CAMERA
- D) ACCESS_FINE_LOCATION

Answer: A

Explanation: SYSTEM_ALERT_WINDOW allows apps to display overlays, often abused by malware.

23.

Which of the following is a key step in penetration testing?

- A) Footprinting
- B) Designing UI
- C) Writing user manuals
- D) Installing software

Answer: A

Explanation: Footprinting gathers info about a target, a crucial first step in penetration testing.

24.

What does certificate pinning do?

- A) Restricts apps to only trust specific SSL certificates
- B) Speeds up downloads
- C) Updates certificates automatically
- D) Disables SSL encryption

Answer: A

Explanation: Pinning limits accepted certificates to trusted ones, blocking MITM.

25.

What kind of attack tries to steal session cookies?

- A) Session hijacking
- B) Buffer overflow
- C) SQL injection
- D) XSS

Answer: A

Explanation: Session hijacking captures cookies to impersonate users.

26.

Which scanning methodology detects open ports stealthily?

- A) Stealth Scan (SYN scan)
- B) Full Connect Scan
- C) Ping Sweep

D) UDP Scan

Answer: A

Explanation: SYN scans do not complete TCP connections, reducing detection risk.

27.

How do attackers use 'banner grabbing'?

A) To gather software version info from servers

B) To steal user credentials

C) To spoof IP addresses

D) To flood networks

Answer: A

Explanation: Banner grabbing collects server info useful for identifying vulnerabilities.

28.

What is the purpose of a honeypot?

A) To lure attackers and study their behavior

B) To speed up network traffic

C) To encrypt data

D) To update antivirus

Answer: A

Explanation: Honeypots are decoy systems designed to attract and monitor attackers.

29.

Which Android security model feature limits app capabilities?

A) Sandbox

B) Root access

C) Overlays

D) Widgets

Answer: A

Explanation: Sandboxing isolates apps to prevent them from accessing each other's data.

30.

Which malware type disguises itself as legitimate software?

- A) Trojan
- B) Worm
- C) Virus
- D) Rootkit

Answer: A

Explanation: Trojans masquerade as useful programs but carry malicious payloads.

☒ Hard (10 Questions)

31.

In an Android WebView, what is a secure practice to prevent XSS attacks?

- A) Disable JavaScript unless necessary and validate inputs
- B) Enable all plugins
- C) Use default WebView settings
- D) Allow third-party cookies

Answer: A

Explanation: Disabling JavaScript reduces attack surface; input validation prevents script injection.

32.

Which Android permission could be exploited to perform Man-in-the-Browser attacks?

- A) BIND_ACCESSIBILITY_SERVICE
- B) INTERNET
- C) CAMERA

D) READ_CONTACTS

Answer: A

Explanation: Accessibility services can be misused to intercept user inputs and manipulate apps.

33.

What is a 'Rogue Access Point' detection technique?

- A) Monitoring for unknown MAC addresses in Wi-Fi logs
- B) Disabling Wi-Fi on all devices
- C) Turning off Bluetooth
- D) Using only cellular data

Answer: A

Explanation: Unknown MAC addresses or unexpected APs indicate rogue access points.

34.

What is the main difference between a worm and a virus?

- A) Worms self-replicate without user action; viruses require a host file
- B) Viruses self-replicate; worms do not
- C) Worms only infect emails; viruses infect files
- D) No difference

Answer: A

Explanation: Worms spread independently, viruses need to attach to files.

35.

Which attack uses crafted TCP flags like FIN, NULL, or XMAS to evade detection?

- A) Stealth scanning
- B) SYN flooding
- C) Buffer overflow
- D) SQL injection

Answer: A

Explanation: These scans exploit TCP flag combinations to probe ports while avoiding logging.

36.

How can DNSSEC help in network security?

- A) By digitally signing DNS data to prevent spoofing
- B) By encrypting all DNS queries
- C) By blocking DNS traffic
- D) By speeding up DNS resolution

Answer: A

Explanation: DNSSEC adds cryptographic signatures to DNS responses to verify authenticity.

37.

Which tool can be used on Android to detect and analyze network sniffing?

- A) Wireshark (via rooted device) or tcpdump
- B) Calculator
- C) File Manager
- D) Gallery

Answer: A

Explanation: Wireshark/tcpdump capture network packets for analysis, requiring root on Android.

38.

Which method prevents smishing attacks?

- A) Do not click suspicious SMS links and verify sender
- B) Use only Wi-Fi networks
- C) Disable Bluetooth
- D) Use antivirus only

Answer: A

Explanation: Awareness and verification prevent falling for SMS phishing scams.

39.

What is a countermeasure to SMB relay attacks?

- A) Use SMB signing and disable NTLMv1 authentication
- B) Disable Wi-Fi
- C) Use VPN
- D) Update the browser

Answer: A

Explanation: SMB signing prevents tampering; NTLMv1 is vulnerable to relay attacks.

40.

Which is NOT a typical symptom of a Trojan infection on Android?

- A) Rapid battery drain
- B) Increased app crashes
- C) Improved device performance
- D) Unusual data usage

Answer: C

Explanation: Trojans degrade performance; improved performance is unrelated.